

# : Final Report :

## AE461 - Aircraft Design

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## Questions

### • Mission Requirements :

|                                                     |
|-----------------------------------------------------|
| Endurance := 500 min $\equiv$ 8 hrs 20 min          |
| Range := 25Km                                       |
| Payload Weight := 735Kg                             |
| Total Passenger := 7 $\equiv$ 5 passengers + 2 crew |

### - Reference Aircraft:

|                    |                                                                                                                            |
|--------------------|----------------------------------------------------------------------------------------------------------------------------|
| Primary Reference  | DA62[1]                                                                                                                    |
| Adjunct References | Cessna 425[2]<br>Cessna 441 Conquest II[3]<br>Mitsubishi MU-2[4]<br>Piper PA31T-Cheyenne[5]<br>Beechcraft King Air 260 [6] |

- Our model will be a *Twin turboprop* with body made up of composite materials.
- The proposed aircraft can serve as chartered aircraft, private aircraft for intercity or tourist planes.

## 1.1 Weight Estimation

**Ans:** We have referred Chapter3 of Raymer for weight estimation. A detailed exposition for the same could be found in Assignment2 (on "Weight Estimation"). Over here only the results are stated :

|                                                 |        |
|-------------------------------------------------|--------|
| Empty weight of the Aircraft, $W_E$             | 2552Kg |
| Fuel weight , $W_f$                             | 901Kg  |
| Payload weight, $W_p$                           | 735Kg  |
| Takeoff weight, $W_{TO} \equiv W_E + W_f + W_p$ | 4158Kg |

### • For the weight estimation calculation we have assumed $\frac{L}{D} = 16$

## 1.2 Power Plant Description

Since our total weight came out to be 4158kg which is equal to be  $4158 * 9.8 = 40748.4N = 9160.6047lbf$ . and our total thrust should be greater than this value.

In the preliminary design phase of a turboprop aircraft, the maximum take-off weight is determined to be 4158 kg and the aircraft  $C_{D_o}$  is determined to be 0.018. By using the matching plot technique, we can determine engine power (P) of the aircraft that is required to have performance capabilities:

### 1. Stall Speed

$$\frac{W}{S} = \frac{\rho v^2 C_{l_{max}}}{2}$$

where  $C_{l_{max}} = 1.3$  and  $V_s = \frac{V_{Emax}}{1.2}$

after putting all the values we get equation as

$$\frac{W}{S} = 3.5187 * 10^3 N/m^2 \equiv 351.091 Kg/m^2 \quad (1)$$

$$\left(\frac{W}{S}\right)_{max} = 3.5187 * 10^3 N/m^2 \equiv 351.091 Kg/m^2$$

## 2. Maximum Speed

The wing and engine sizing based on maximum speed requirements for a prop-driven aircraft are represented by equation

$$\left(\frac{W}{P_{SL}}\right)_{V_{max}} = \frac{\eta_p}{0.5\rho_o V_{max}^3 C_{D_0} \frac{1}{S} + \frac{2KW}{\rho\sigma V_{max} S}}$$

after putting all the known values we get equation as

$$\frac{W}{P} = \frac{0.8}{\frac{7389.144}{\frac{W}{S}} + 0.0029 \frac{W}{S}} \quad (2)$$

## 3. Take Off Run

The wing and engine sizing based on take-off run requirements for a prop-driven aircraft are represented by equation

$$\frac{W}{P} = \frac{1 - \exp(0.6\rho * g * C_{D_G} S_{T_0} \frac{1}{S}) \frac{\eta_p}{V_{T_0}}}{\mu - (\mu + \frac{C_{D_G}}{C_{L_R}})(\exp(0.6\rho g C_{D_G} S_{T_0} \frac{1}{S}))}$$

after putting all known values we get above equation as

$$\frac{W}{P} = \frac{1 - \exp(7.6801 \frac{1}{S})}{0.04 - ((0.0967)(\exp(7.6801 \frac{1}{S})))} \quad (3)$$

## 4. Service ceiling

The wing and engine sizing based on service ceiling requirements for a prop-driven aircraft are represented by equation

$$\frac{W}{P} = \frac{\sigma_c}{\frac{(ROC)_c}{\eta_P} + \sqrt{\frac{2 \frac{W}{S}}{\rho \sqrt{\frac{3C_{D_0}}{K}}} (\frac{1.155}{(\frac{L}{D})_{max} \eta_P})}}$$

after putting all known values we get above equation as

$$\frac{W}{P} = \frac{0.2546}{0.635 + (0.1207 \sqrt{\frac{W}{S}})} \quad (4)$$

## 5. Rate of Climb

The wing and engine sizing based on rate of climb requirements for a prop-driven aircraft are represented by equation

$$\frac{W}{P} = \frac{1}{\frac{ROC}{\eta_P} + \sqrt{\frac{2}{\rho \sqrt{\frac{3C_{D_0}}{K}}} * \frac{W}{S} (\frac{1.155}{(\frac{L}{D})_{max} \eta_P})}}$$

after putting all known values we get above equation as

$$\left(\frac{W}{P}\right) = \frac{1}{0.635 + (0.1207 \sqrt{\frac{W}{S}})} \quad (5)$$

Now if we plot (1), (2), (3), (5) and(4) on a plot then we get the following figure. here you can see red plot for Stall Speed, blue for Maximum Speed, Green for Take Off Run , Violet for Service Ceiling , Black for Rate of Climbing.

1. Region above the curve corresponding to maximum speed is acceptable.
2. Region below the curve corresponding to take-off distance is acceptable.
3. Region below the curve corresponding to ROC is acceptable.

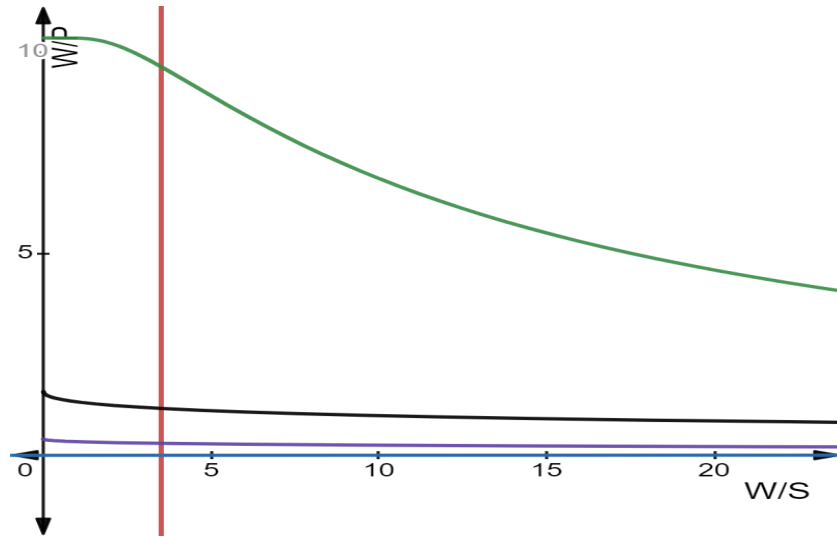


Figure 1: Matching plot for power

4. Region above the curve corresponding to ceiling is acceptable.
5. Region on the left of the curve corresponding to stall speed is acceptable.

so

$$\frac{W}{P} = value1$$

$$\frac{W}{S} = value2$$

the region below each graph is satisfying the performance requirements. We are looking for the smallest engine (lowest power) that has the lowest operating cost. we choose as

$$\frac{W}{P} = 0.025$$

$$P = \frac{W_{T0}}{\frac{W}{P}}$$

$$P = \frac{4158}{0.025} = 166.320KW$$

In this way we decided power plant based on required power. Please refer the code link for calculation and plot.

- We have the following two commercially available turboprop engines, which satisfies our requirement :

1. **Pratt & Whitney Canada PT6-114**

This engine has max Power 600shp  $\equiv$  441.229Kw

2. **Honeywell TPE331-25B**

This engine has max Power 620shp  $\equiv$  456.009Kw

shp  $\rightarrow$  Shaft Horse Power

### 1.3 Wing design

1. *Wing Loading,  $\frac{W}{S}$*

$$\frac{W}{S} = 190Kg/m^2 \quad (6)$$

It is decided after going through a list of aircrafts which have similar mission requirements

## 2. Wing Area, $S$

$$S = \frac{W}{\text{Wing Loading}} \implies S = 21.8842\text{m}^2 \quad (7)$$

## 3. Aspect Ratio, $\mathcal{R}$

$$\mathcal{R} = 10.22 \equiv \left(\frac{9.2}{0.9}\right) \quad (8)$$

From the table we conclude that a Twin turboprop has an  $\mathcal{R} = 10.22$ . Further we need to divide it by a factor  $0.75 - 0.9$ . I have selected a factor of  $0.9$ .

**Table 4.1 Aspect ratio**

| Sailplane equivalent* aspect ratio = 4.464 (best $L/D$ )* <sup>69</sup> |                                           |        |
|-------------------------------------------------------------------------|-------------------------------------------|--------|
| Propeller aircraft                                                      | Equivalent aspect ratio                   |        |
| Homebuilt                                                               | 6.0                                       |        |
| General aviation—single engine                                          | 7.6                                       |        |
| General aviation—twin engine                                            | 7.8                                       |        |
| Agricultural aircraft                                                   | 7.5                                       |        |
| Twin turboprop                                                          | 9.2                                       |        |
| Flying boat                                                             | 8.0                                       |        |
| Jet aircraft                                                            | Equivalent aspect Ratio = $aM_{\infty}^2$ |        |
|                                                                         | $a$                                       | $C$    |
| Jet trainer                                                             | 4.737                                     | -0.979 |
| Jet fighter (dogfighter)                                                | 5.416                                     | -0.622 |
| Jet fighter (other)                                                     | 4.110                                     | 0.622  |
| Military cargo/bomber                                                   | 5.570                                     | 1.075  |
| Jet transport                                                           | 7.50                                      | 0      |

\*Equivalent aspect ratio = wing span squared/(wing and canard areas)

Figure 2: Aspect ratio for different type of aircraft [7]

## 4. Wing Span, $b$

$$b = \sqrt{\mathcal{R}S} = 14.96\text{m} \quad (9)$$

## 5. Wing sweep, $\Lambda_{LE}$ and $\Lambda_{\frac{c}{4}}$ $\Lambda_{LE} \rightarrow$ Leading Edge sweep angle

$\Lambda_{\frac{c}{4}} \rightarrow$  Quarter chord sweep angle

Our plane will fly in the subsonic regime only therefore introducing any wing sweep will not add onto any benefits apart from may be aesthetics. To simplify my design I will consider  $\Lambda_{\frac{c}{4}} = 0$  deg.

$$\tan \Lambda_{LE} = \tan \Lambda_{\frac{c}{4}} + \frac{1 - \lambda}{\mathcal{R}(1 + \lambda)} \quad (10)$$

using (10) we get  $\Lambda_{\frac{c}{4}} = 2.4013$  deg

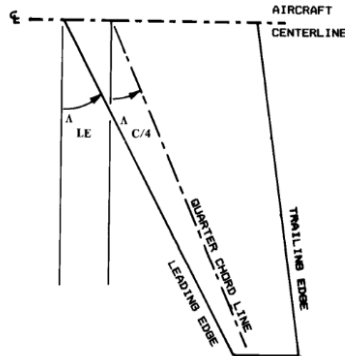


Figure 3: Description of Wing sweep[7]

#### 6. Taper Ratio, $\lambda$

$$\lambda = \frac{C_{root}}{C_{tip}} \quad (11)$$

As mentioned in [7]

|                   |                       |
|-------------------|-----------------------|
| For straight wing | $\lambda = 0.4 - 0.5$ |
| For swept wing    | $\lambda = 0.2 - 0.3$ |

Therefore **we have selected**  $\lambda = 0.4$

#### 7. Chord Length

$$C_{root} = \frac{2S}{b(1+\lambda)} = 2.0898m \quad (12a)$$

$$C_{tip} = \lambda C_{root} = 0.8359m \quad (12b)$$

$$\bar{C} = \frac{2}{3} C_{root} \frac{1+\lambda+\lambda^2}{1+\lambda} = 1.5524m \quad (12c)$$

$$\bar{Y} = \frac{b}{6} [(1+\lambda)(1+2\lambda)] = 6.2832m \quad (12d)$$

$C_{root} \rightarrow$  Root chord,  $C_{tip} \rightarrow$  Tip chord,  $C_{MAC} \rightarrow$  Mean Aerodynamic chord and  $\bar{Y} \rightarrow$  Location of the mean Aerodynamic chord

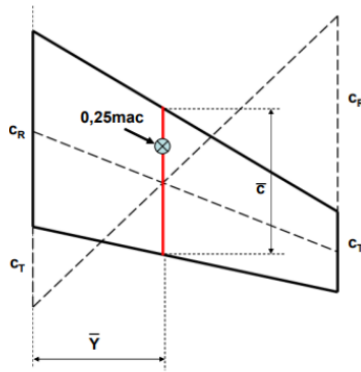


Figure 4: Description of chords of typical wing [7]

#### 8. Dihedral Angle, $\phi$

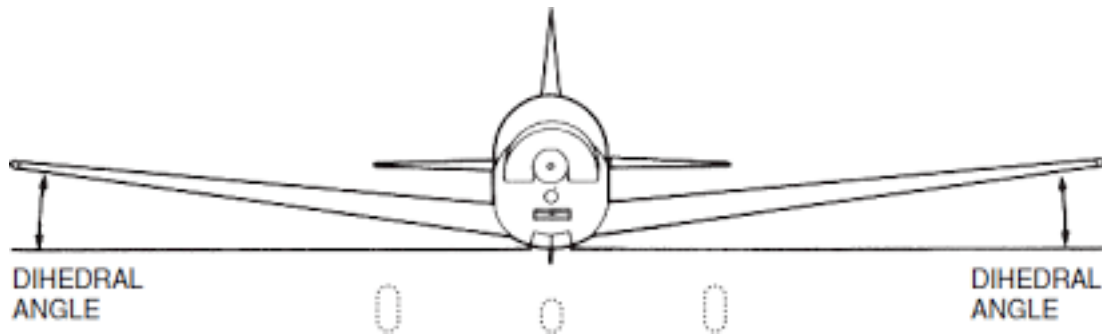


Figure 5: Dihedral Angle [8]

Our aircraft has a low wing configuration, therefore we have selected  $\phi = 5.5 \text{ deg}$

|                  | Wing position |        |         |
|------------------|---------------|--------|---------|
|                  | low           | mid    | high    |
| Unswept          | 5 ÷ 7         | 2 ÷ 4  | 0 ÷ 2   |
| Subsonic swept   | 3 ÷ 7         | -2 ÷ 2 | -5 ÷ -2 |
| Supersonic swept | 0 ÷ 5         | -5 ÷ 0 | -5 ÷ 0  |

Figure 6: Dihedral Angle of typical wing configuration [7]



Figure 7: Wing twist [7]

#### 9. *Twist, $\theta$*

Our aircraft has  $\theta = 3$  deg

- **Airfoil Selection** : We have followed an iterative method for Airfoil selection of wing.

Step 1: We decided that a 4 digit airfoil will be used.

We varied max camber and max thickness to come up with a suitable airfoil.

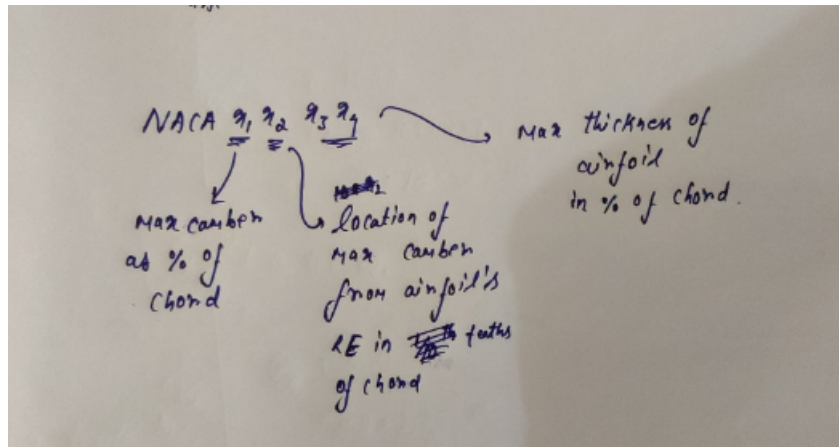


Figure 8: NACA 4 digit Airfoils

Step 2: our model aircraft will have  $C_{L,max} = 1.3$

in the [9] website we set the following setting to obtain the set of valid airfoils:

|                       |      |
|-----------------------|------|
| Min Airfoil camber    | 0%   |
| Max Airfoil camber    | 3.5% |
| Min Airfoil thickness | 8%   |
| Max Airfoil thickness | 13%  |

There are only five such airfoils NACA 2.5411, NACA 2412, NACA 2410, NACA 1410 and NACA 1412

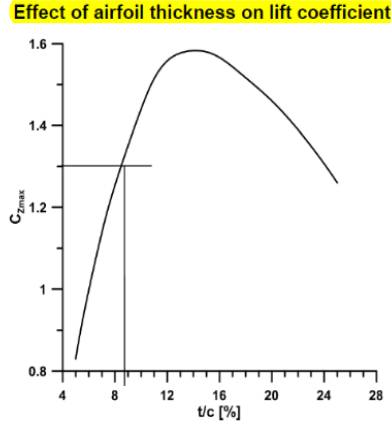


Figure 9:  $C_{L,max}$  and Airfoil Thickness

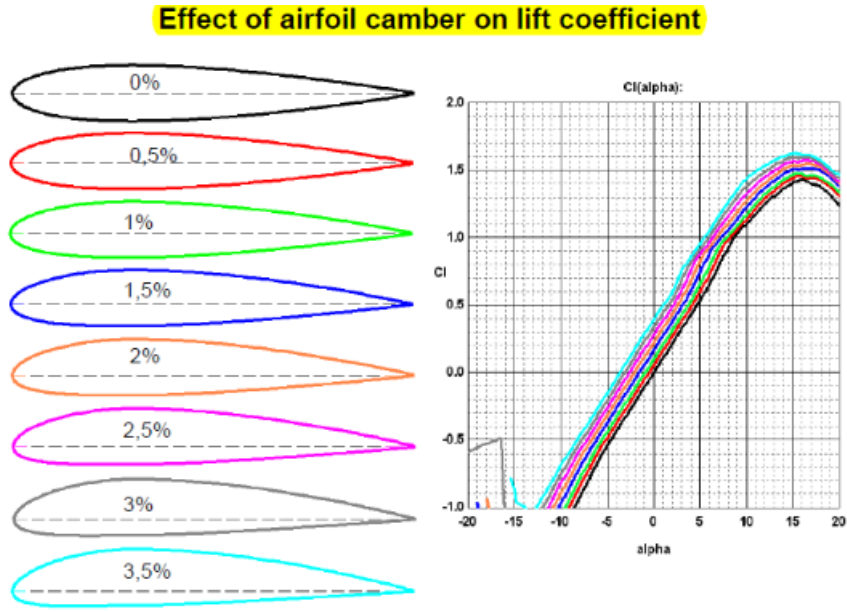


Figure 10:  $C_L$  and Airfoil Camber

Step 3: We have the following necessary condition to be satisfied by any airfoil

$$C_{l_\alpha} = \frac{C_{l_{max}}}{\alpha_{stall}^{2D} - \alpha_{L=0}} \quad (13a)$$

$$C_{L_\alpha} = \frac{C_{L_{max}}}{\alpha_{stall}^{3D} - \alpha_{L=0}} \quad (13b)$$

$$\alpha_{stall}^{3D} \geq \alpha_{stall}^{2D} \quad (13c)$$

Further from finite wing theory:

$$C_{L_\alpha} = \frac{C_{l_\alpha}}{1 + \frac{57.3 C_{l_\alpha}}{\pi \mathcal{R} e}} \quad (14)$$

Here  $e$  is Span efficiency factor.

using (13a), (13b), (13c) and (14) we have the following constraint:

$$\boxed{C_{L_{max}} \left(1 + \frac{57.3 C_{l_\alpha}}{\pi \mathcal{R} e}\right) \geq C_{l_{max}}} \quad (15)$$

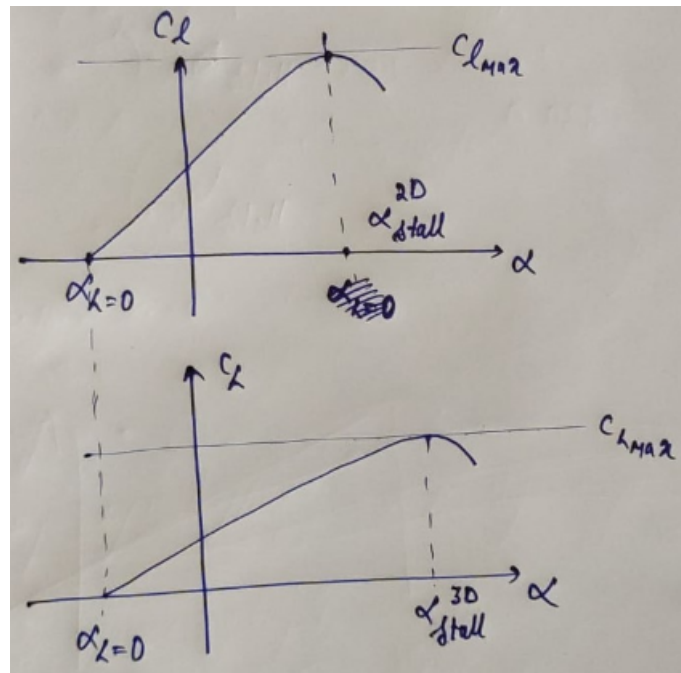


Figure 11:  $C_L$  vs  $C_D$

Out of the five airfoils that we have only **NACA1412** satisfies the above constraint; Hence we consider only **NACA1412**.

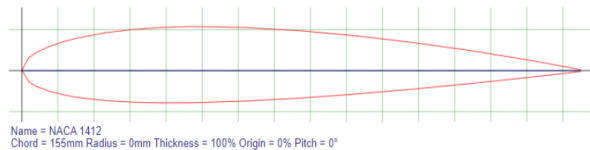


Figure 12: NACA1412 airfoil

### • Variation of force and moment coefficient

#### 1. $C_L$ vs $\alpha$ for the wing

Aerodynamic analysis for our wing and upcoming plots are done by using VSPAERO analysis in VSP software. In VSPAERO analysis it won't account in the viscous effects and flow separation phenomenon, so in the above plot we won't be able to observe the stall characteristics of our wing but we already know that stall occurs at latter angle of attack compared to 2D airfoil.

#### 2. $C_L$ vs $C_D$ for the wing

while Low  $C_D$  is good for going fast and high  $C_L$  is good for heavy lift. Note, however, that a high  $C_L$  requires a higher  $C_D$  due to induced drag component.

#### 3. $C_M$ vs $\alpha$ for the wing

In the below graph we can clearly see that as the angle of attack(AOA) increases the moment coefficient about CG of wing increases which means the wing tries to pitch up similarly as the AOA decreases the wing tries to pitch down which is clearly unstable in the pitch axis (Y-axis for our convention).

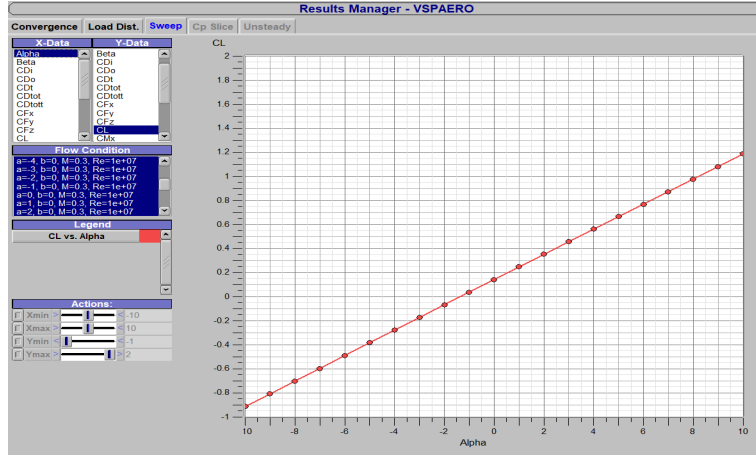


Figure 13:  $C_L$  vs  $\alpha$  for wing

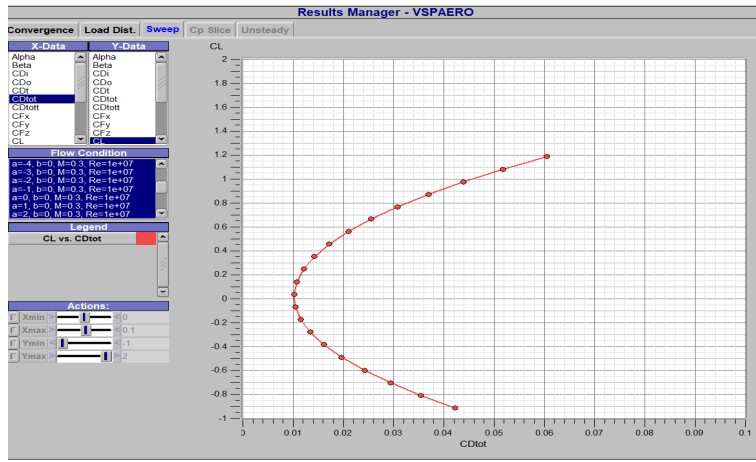


Figure 14:  $C_L$  vs  $C_D$  for wing

## 1.4 Fuselage Design

After comparing fuselage design of many similar aircrafts (refer "Reference Aircraft" list in the first page)

Length of the fuselage,  $L_f = 11m$

|                        |        |
|------------------------|--------|
| Length of Frontal Part | 1.87m  |
| Cockpit Length         | 0.88 m |
| Cabin Length           | 4.84m  |
| Rear Part Length       | 3.41m  |

The Fuselage will have elliptical subsections of dimensions as shown:

## 1.5 Tail design (horizontal and vertical tails)

We are considering a "Conventional" tail design only. The design of the tail would be done using the concept of tail volume coefficients.

$$V_{ht} = \frac{l_{ht} S_{ht}}{S_w c_w} \quad (16a)$$

$$V_{vt} = \frac{l_{vt} S_{vt}}{S_w b_w} \quad (16b)$$

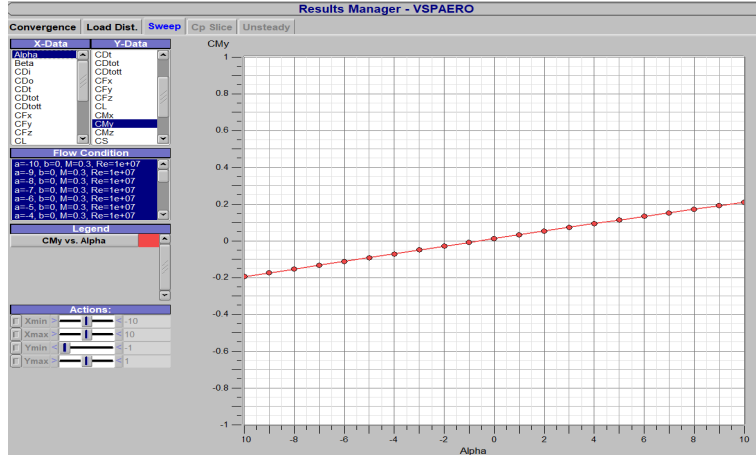


Figure 15:  $C_M$  vs  $\alpha$  for wing

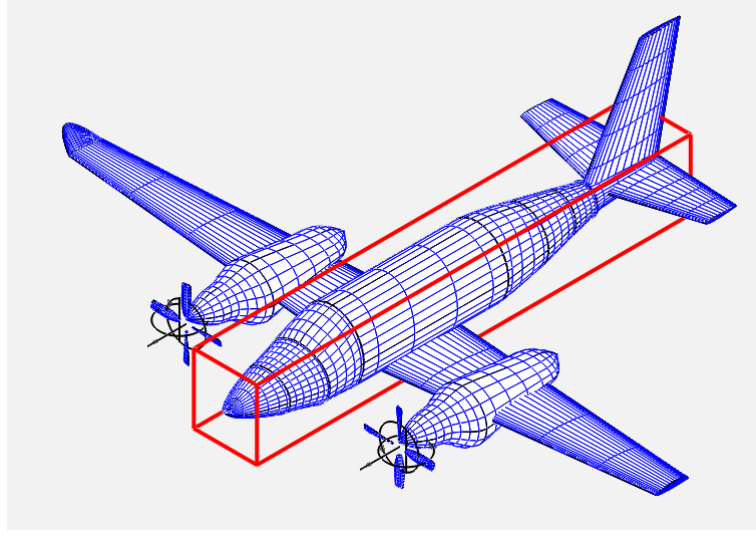


Figure 16: Fuselage of our model aircraft

$v_{ht}$  → Tail volume coefficient of the horizontal tail

$v_{vt}$  → Tail volume coefficient of the vertical tail

$l_{ht}$  → Distance between aerodynamic centers of wing and horizontal tail

$S_{ht}$  → Area of horizontal tail

$l_{vt}$  → Distance between aerodynamic centers of wing and vertical tail

$S_{vt}$  → Area of vertical tail

$S_w$  → Wing area

$C_w$  → Mean Aerodynamic chord of wing

$b_w$  → Span of wing For tail design symmetric airfoils are selected. For our purpose we are using NACA0010 airfoil (both in horizontal as well as vertical tail)

For aircraft with engines mounted on the wing the tail arm is at 50 – 55% of the fuselage length.

### 1.5.1 Horizontal Wing Configuration

Aspect ratio of horizontal tail,  $AR_{ht} = 5$

Taper ratio of horizontal tail,  $\lambda_{ht} = 0.5$

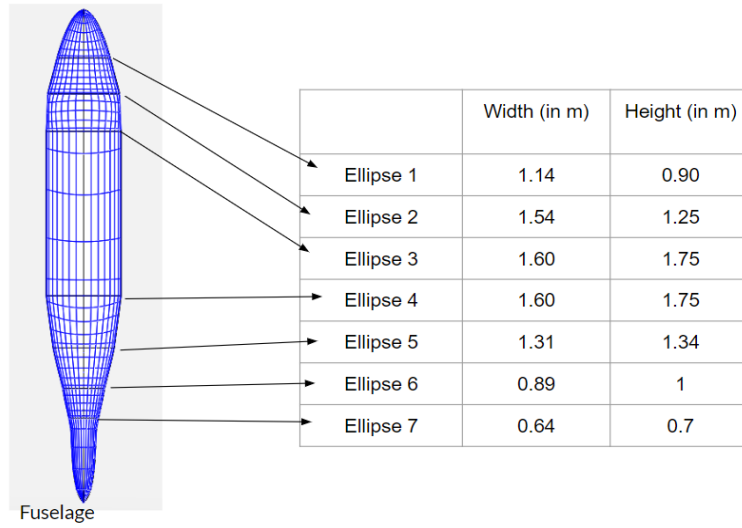


Figure 17: Cross section dimensions of the fuselage

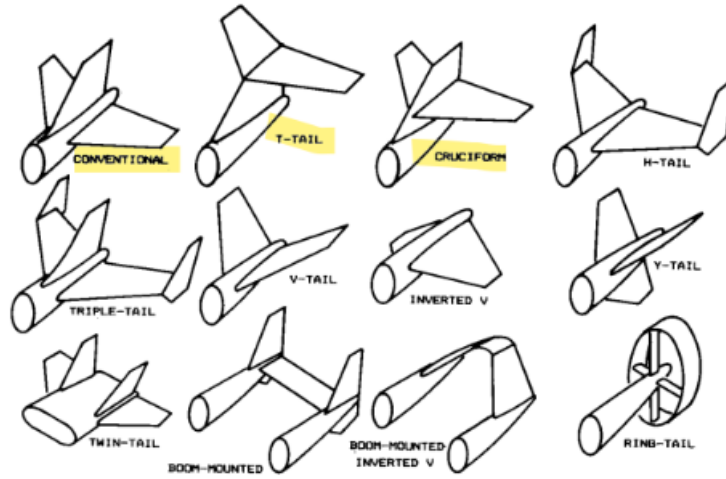


Figure 18: Aft Tail Variations[7]

$$v_{vt} = 0.90$$

$$l_{ht} = 50\% \text{ of } 11 \text{ m} = 5.5 \text{ m}$$

$$S_{ht} = \frac{V_{ht} S_w C_w}{l_{ht}} = 5.56 \text{ m}^2$$

$$b_{ht} = \sqrt{AR_{ht} S_{ht}} = 4.72 \text{ m}$$

Leading edge sweep angle  $\Lambda_{ht} = 5^\circ$  (LE sweep of the horizontal tail is usually set 5 deg more than the wing sweep)

Dihedral Angle =  $5.5^\circ$  (general between  $4^\circ$  to  $6^\circ$ )

| Parameter                | Value              |
|--------------------------|--------------------|
| Aspect Ratio             | 5                  |
| Span                     | 4.72m              |
| Length of tail ( $L_t$ ) | 5.5m               |
| Area of tail ( $S_t$ )   | 5.56m <sup>2</sup> |

| Table 6.4 Tail volume coefficient |                     |                   |
|-----------------------------------|---------------------|-------------------|
|                                   | Typical values      |                   |
|                                   | Horizontal $c_{HT}$ | Vertical $c_{VT}$ |
| Sailplane                         | 0.50                | 0.02              |
| Homebuilt                         | 0.50                | 0.04              |
| General aviation—single engine    | 0.70                | 0.04              |
| General aviation—twin engine      | 0.80                | 0.07              |
| Agricultural                      | 0.50                | 0.04              |
| Twin turboprop                    | 0.90                | 0.08              |
| Flying boat                       | 0.70                | 0.06              |
| Jet trainer                       | 0.70                | 0.06              |
| Jet fighter                       | 0.40                | 0.07              |
| Military cargo/bomber             | 1.00                | 0.08              |
| Jet transport                     | 1.00                | 0.09              |

Figure 19: Tail Volume Coefficients[7]

| Table 4.3 Tail aspect ratio and taper ratio |                 |           |               |           |
|---------------------------------------------|-----------------|-----------|---------------|-----------|
|                                             | Horizontal tail |           | Vertical tail |           |
|                                             | $A$             | $\lambda$ | $A$           | $\lambda$ |
| Fighter                                     | 3–4             | 0.2–0.4   | 0.6–1.4       | 0.2–0.4   |
| Sail plane                                  | 6–10            | 0.3–0.5   | 1.5–2.0       | 0.4–0.6   |
| Others                                      | 3–5             | 0.3–0.6   | 1.3–2.0       | 0.3–0.6   |
| T-Tail                                      | —               | —         | 0.7–1.2       | 0.6–1.0   |

Figure 20: Aspect Ratio and taper ratio for tail[7]

### 1.5.2 Vertical Tail Design

Aspect ratio of horizontal tail,  $AR_{vt} = 1.6$

Taper ratio of horizontal tail,  $\lambda_{vt} = 0.5$

$$v_{vt} = 0.08$$

$$l_{vt} = 50\% \text{ of } 11 \text{ m} = 5.5 \text{ m}$$

$$S_{vt} = \frac{V_{vt} S_w C_w}{l_{ht}} = 4.76 \text{ m}^2$$

$$b_{vt} = \sqrt{AR_{vt} S_{ht}} = 2.76 \text{ m}$$

Quarter chord sweep angle  $\Lambda_{vt,c/4} = 30^\circ$  (Vertical Tail sweep varies between about 35-55 deg. For a low speed aircraft there is little reason why vertical tail sweep should be beyond 20 deg other than the aesthetics)

| Parameter                | Value             |
|--------------------------|-------------------|
| Aspect Ratio             | 1.6               |
| Span                     | 2.76m             |
| Length of tail ( $L_t$ ) | 5.5m              |
| Area of tail ( $S_t$ )   | 2.76 $\text{m}^2$ |

### • Variation of moment coefficient

#### 1. Stability along pitch and Yaw axes

In the wing section we saw that the slope of  $C_M$  curve ( $C_{M_\alpha}$ ) has positive value which is clearly unstable so to make our aircraft stable we have to move CG of aircraft to a desired location where our aircraft is actually stable along pitch axis and yaw axis which means negative slopes ( $C_{M_\alpha}$  and  $C_{M_\beta}$ ). To find the desired location of CG, we can keep on changing the reference point at which moments are taken and finding out at which point we are getting negative values for both  $C_{M_\alpha}$  and  $C_{M_\beta}$ . This reference point is

our desired CG. our desired CG location is the Red dot as shown in the figures below ( $x=1,y=0,z=0.6$ )

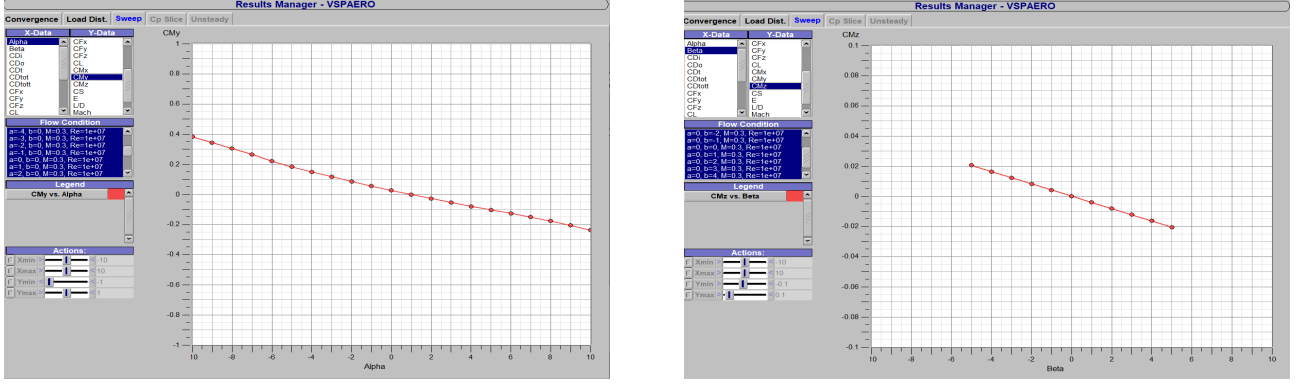


Figure 21:

LEFT:  $C_M$ (pitching moment) vs  $\alpha$

RIGHT:  $C_M$ (yaw moment) vs  $\beta$

so to make our aircraft stable we keep the engine at a location such that CG of entire aircraft coincides with the desired CG.

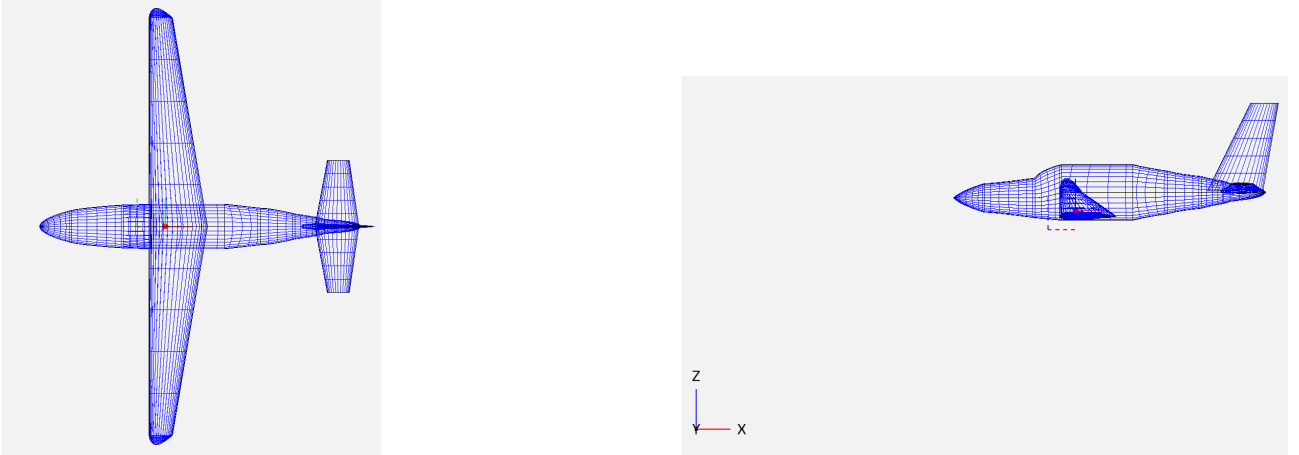


Figure 22: Desired CG (red point)

LEFT: Top View

RIGHT: Side View

## 1.6 Vn diagram and gust Vn diagram

To plot the structural operating envelope, we need the following information:

- $W_T = 4158 \text{ kg}$
- $S = 21.8842 \text{ m}^2$
- $C_{z,max} = 1.6$  (using  $1.25 C_{z,max}$  relation) and  $C_{z,min} = -1.0$
- Load factor ( $n$ ) is given by Lift/Weight ratio. It can be represented with the equation:  $n = \frac{C_z \rho S V^2}{2W}$ . Using

the values of  $C_{z,max}$  and  $C_{z,min}$  we get the equation that relates velocity with load factor.

- For **Gust V-n Diagram**, we use the following relation:

-  $n_g = 1 \pm \frac{kUaV}{498W/S}$  in imperial units. Here we take  $kU = 17 \text{ m/s}$  (from FAR-25 regulations for sea-level calculations)

$$-n_g = 1 \pm 0.027V$$

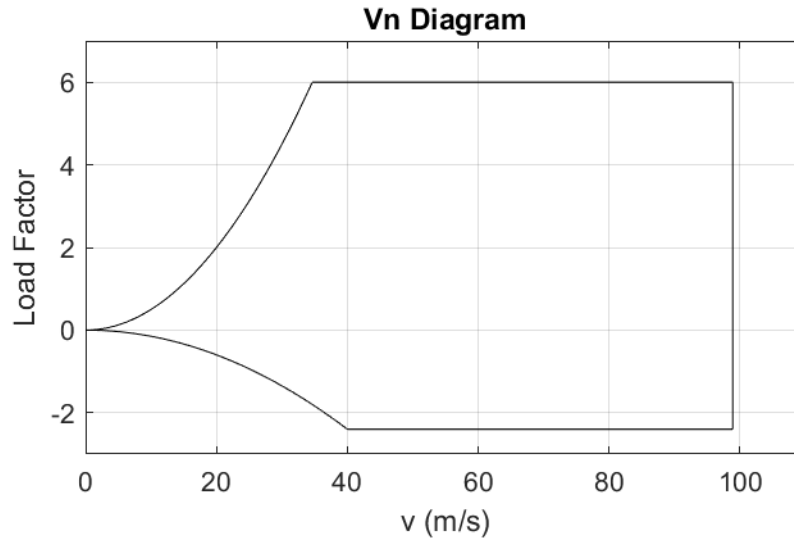


Figure 23: Vn Diagram

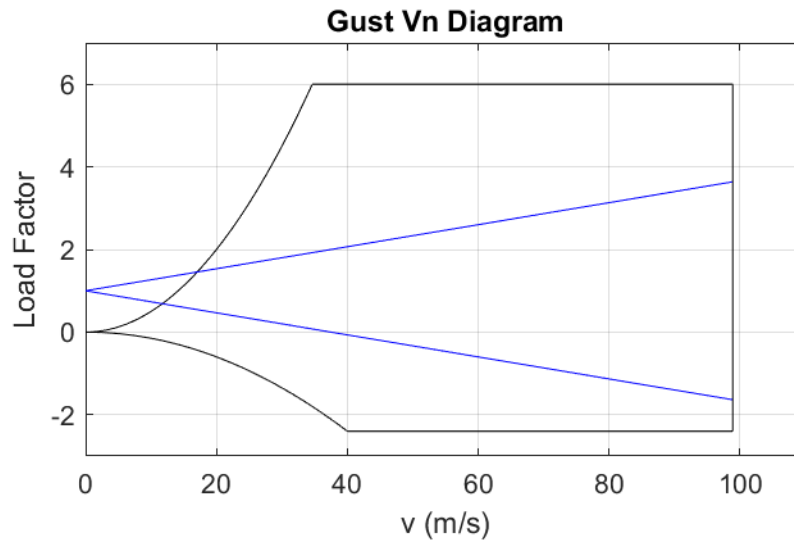


Figure 24: Combined V-n Diagram and Gust V-n Diagram

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